Habit Mastery League

Project report Mobile Application Development, Project 1,

undergraduate Abubeker Mohammed and Jalen Artis

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## What we built

Our app is called Habit Mastery League. It help a student keep a weekly habit, like studying, reading, or drinking water. They add the habit, mark the days they finish, and look at the history later. It work without internet. We did not use a cloud database.

We have five screens. Home is the habit list. Add and Edit is one screen, and you open it from Home. Check-In is for marking a day. History show the old check-ins, a category filter, and the completion rate. Settings save a name and dark mode on the phone. The bottom menu is Home, Check-In, History, and Settings.

SQLite has two tables, habits and check-ins. A habit store the name, category, weekly goal, and if the week is running. A check-in store the date, if they did it, and a short note. SharedPreferences is only for the name, dark mode, and a small flag so the sample habits do not come back after you delete them.

The goal card is on the Add/Edit screen. It look at the check-ins from the last 7 days and suggest a weekly goal. This is a rule we wrote, not a trained model. The card say it is suggested, not guaranteed. Apply only move the slider. Not now hide the card. Nothing is saved until the user tap Save. We also show a fixed starter goal of 3 days so we can compare. More about that is in AI_FEATURE.md.

Who did what

Abubeker did Home, Add/Edit, the SQLite tables, and a lot of the writing. Jalen did Check-In, History, Settings, and the goal helper. We both tested. I used my Samsung phone. Jalen ran the app on an emulator on his PC and said it worked.

What went wrong

The big problem was my laptop. It only have 8 GB of RAM. I had VS Code open to write the code, and I also opened Android Studio so I can start the emulator. The emulator already need a lot of memory. When VS Code, Android Studio, and the emulator was all open, the laptop freeze. One time the build could not even read the files, and Gradle just waited until it timed out. Android Studio was still holding the project, so the next build got stuck on that lock.

I closed Android Studio after the emulator opened, but the emulator was still too heavy for this laptop. So I stopped using it. I plugged in my Samsung phone and turned on USB debugging, and I ran the app there. That was better for the RAM. We still write the code in VS Code. Jalen used an emulator on his own PC, so my laptop did not have to run one. Another problem was saving a check-in for the same habit on the same day. If the user tap save again, we would get two rows for one day, and the completion rate would be wrong. Now we do not add the second row. If that day already have a check-in, the button say Update check-in and it change the old one.

Deleting a habit was also easy to get wrong. If the habit is gone but the check-ins stay, History still show them and they are not connected to a habit. So we delete

the check-ins in the same database step. The table also have ON DELETE CASCADE. We tested that a parent delete does not leave old rows.

The goal card can go up too fast. If there is only one or two check-ins and they are both done, the rule raise the weekly goal. That can be too high for a new habit. The card say the sample is small and the suggestion can be too high. The goal also stay between 1 and 7. If it is already 7, it stay at 7. The five cases we checked are a normal week, no check-ins, a bad goal, the goal already at 7, and that short lucky streak.

In this app a week is the last 7 days, not Monday to Sunday. We did it this way because it was easier to count. It is not a real calendar week, so we say that on the Settings screen.

The first picture we tried to take was black. The phone was on the charging screen, so the app was not really showing. We woke the phone and took the pictures again. They are in the phone_screens folder.

When the input is bad

If the habit name is empty, it does not save. The form say please enter a habit name. If the name is longer than 40 letters, that is stopped too. The weekly goal is a slider from 1 to 7, so the user cannot put 0 or 8. If there is no check-ins yet, the goal card do not make up a number from history. It say there is not enough data, and it show the fixed starter goal of 3.

On Check-In, the date picker stop at today, so you cannot pick a future day. A note longer than 120 letters is not saved. If that habit already have a check-in for the

day, the app do not add a second one. It tell the user and update the old row. Delete and reset ask first, so one tap do not wipe the habit or the last 7 days.

If the list cannot load, the screen say it could not load and there is a Try again button.

It do not just go blank. If the database cannot open when the app start, it say to close the app and try again. If there is no habits, Home say to tap Add habit instead of showing nothing.

How we checked it

We ran flutter test for the goal rule and for deleting a habit. On my Samsung phone I opened Home, Check-In, History, Settings, Edit habit, and dark mode. The sample habits was still there after I left the app and came back. Jalen ran the project on an emulator on his PC and said it worked. The file the class want is the release APK, app-release.apk.

On October 31, I ran flutter analyze and flutter test again. Flutter analyze showed no issues, and all 17 tests passed. I also tested editing check-ins in History on my Samsung phone. I saved a changed note, canceled another edit, and opened and closed the edit window several times. Everything worked without errors. After that, I built a new release APK with the latest changes.